

GOVOS

MASTER PROJECT BLUEPRINT

Administrative • Product • Workflow • Architecture • Security • Implementation

The entire GovOS concept on one page — expanded in the following pages.

Use of this edition: this is a visual planning and explanation document. It preserves the source concept while distinguishing source-derived decisions from recommendations and validation items.

1. Executive Overview

GovOS is a government operational workflow and visibility platform centered on citizen/public issues, departmental execution, SLA management, verified outcomes, auditability, tenant isolation and leadership monitoring.

- Every work item should have an accountable owner.
- Time-bound execution is represented through SLA monitoring and escalation.
- Completion and verification are separate events.
- Leadership manages exceptions and outcomes rather than individual records.

2. GovOS Vision

The target is a traceable government work lifecycle: need → case → ownership → execution → verification → outcome → management insight.

3. Problem Statement

GovOS addresses fragmented operational visibility, unclear ownership, weak cross-department coordination, passive SLA reporting and the risk of treating reported completion as verified outcome.

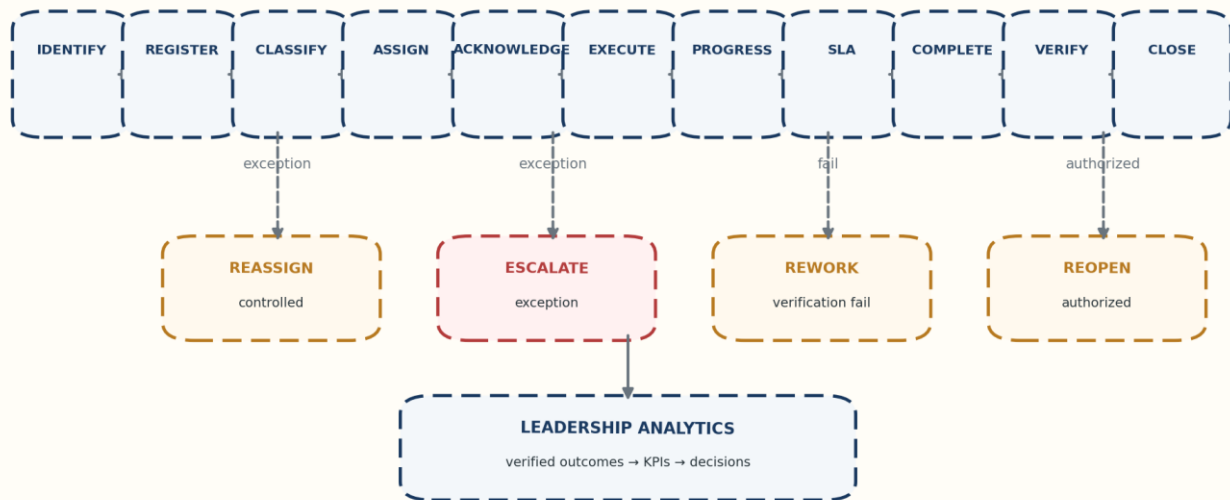
4. Operating Model

The operating model is built around single ownership, time-bound execution, verified outcomes, auditability, tenant isolation, exception management, interoperability and phase discipline.

5. Master Workflow

01. MASTER GOVOS LIFECYCLE

The complete government work lifecycle

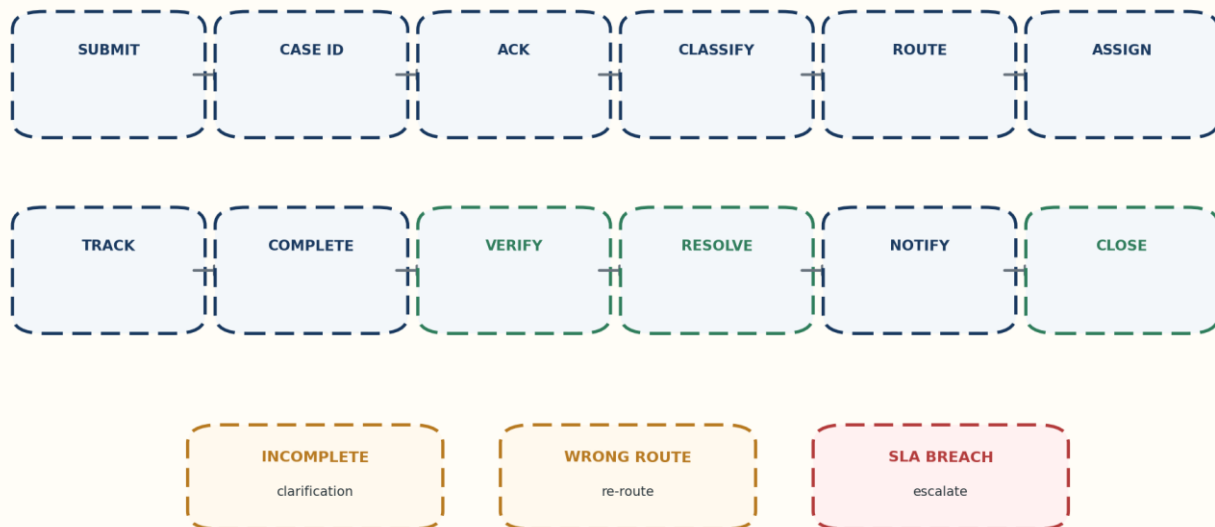


Master lifecycle and controlled exception paths.

6. Citizen / Public Issue Workflow

02. CITIZEN / PUBLIC ISSUE WORKFLOW

Simple citizen experience; complex administration stays behind the service

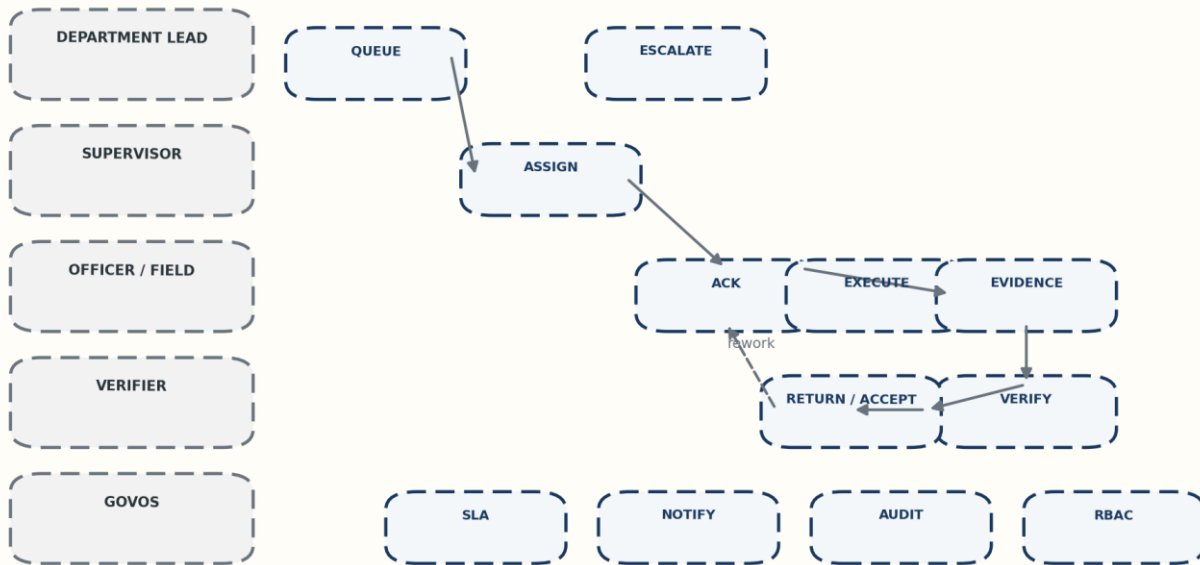


Citizen-facing journey.

7. Department & Officer Workflow

03. DEPARTMENT SWIMLANE

Who does what — and what the platform records

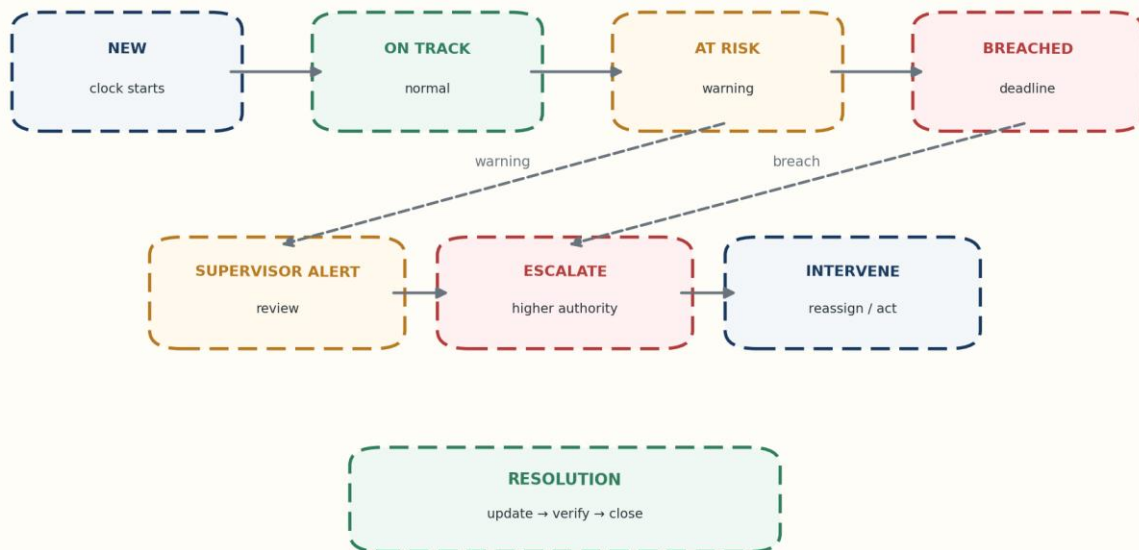


Department swimlane.

8. SLA, Escalation & Exception

04. SLA + ESCALATION

Exact values must be configured and validated for the pilot

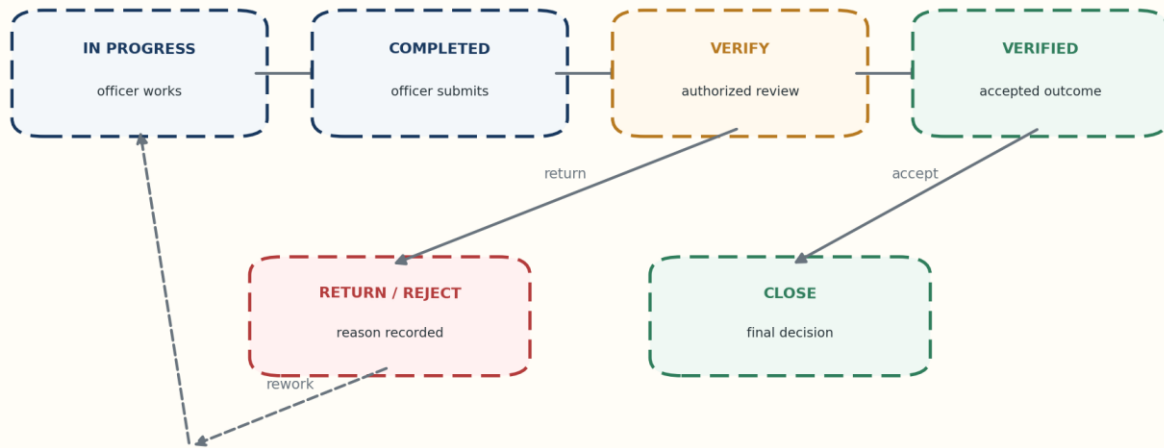


SLA state machine; exact values require pilot validation.

9. Completion & Verification

05. COMPLETION ≠ VERIFICATION

A core GovOS governance control

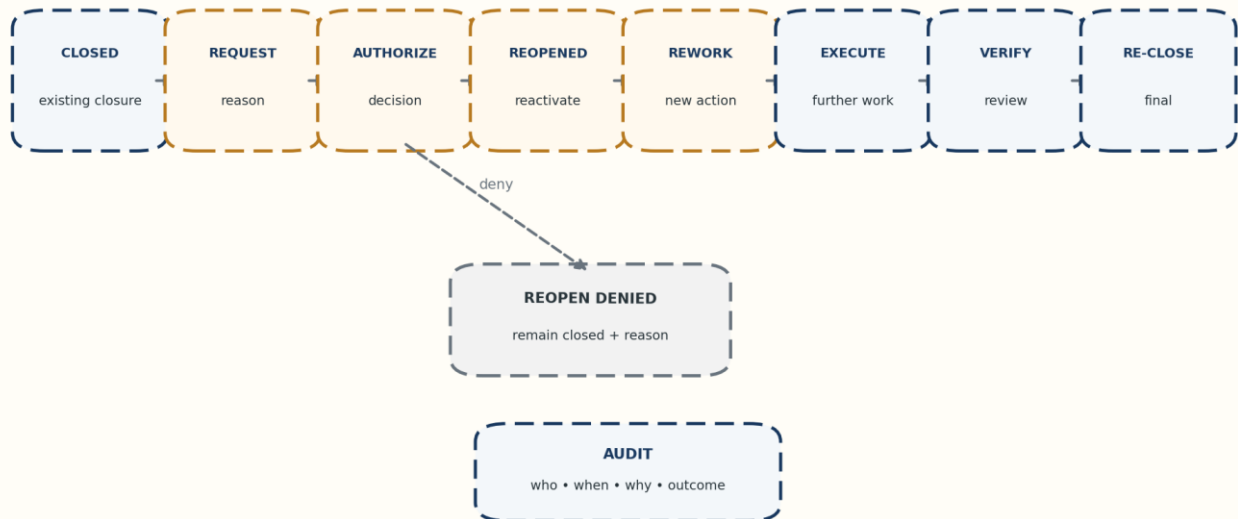


Completion is not verification.

10. Reopen / Rework

06. REOPEN / REWORK

Controlled re-entry into execution

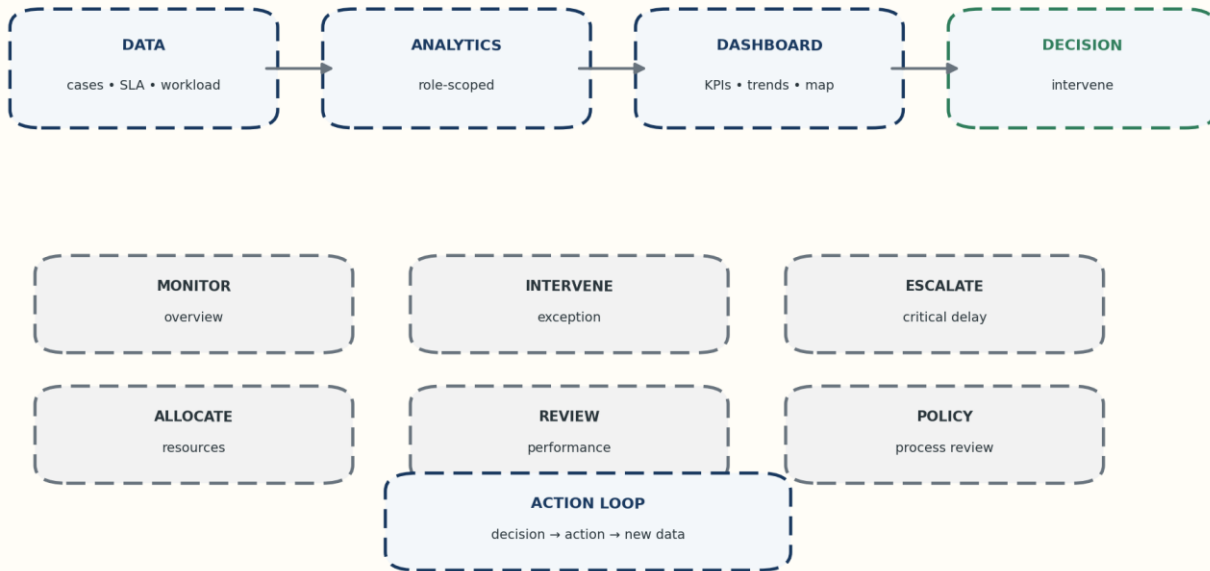


Controlled re-entry after closure.

11. Leadership / Command Centre

07. LEADERSHIP / COMMAND CENTRE

Exception-based management

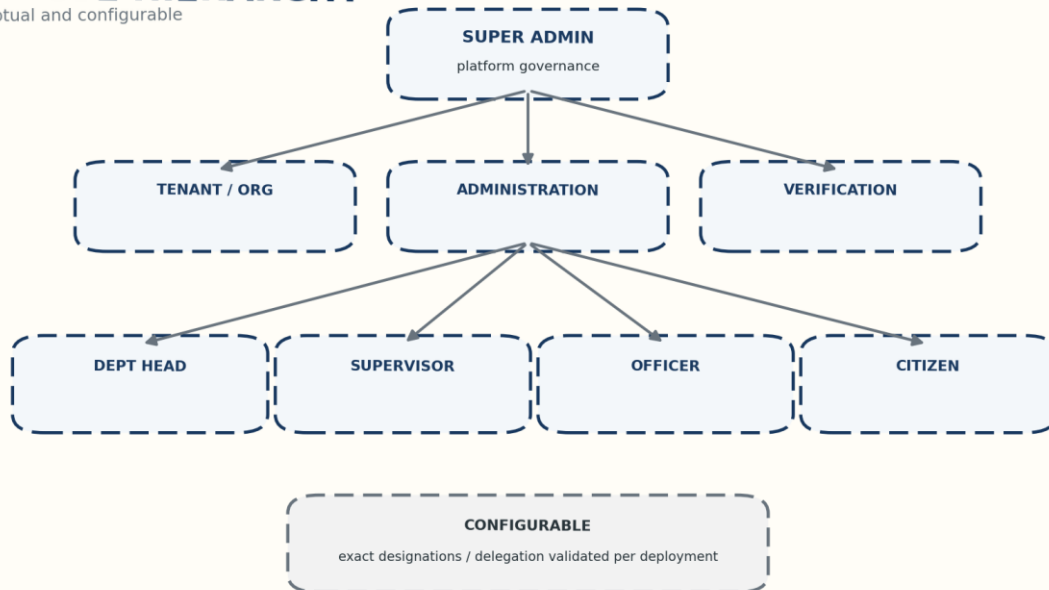


Exception-based leadership model.

12. Role & Responsibility Model

08. ROLE HIERARCHY

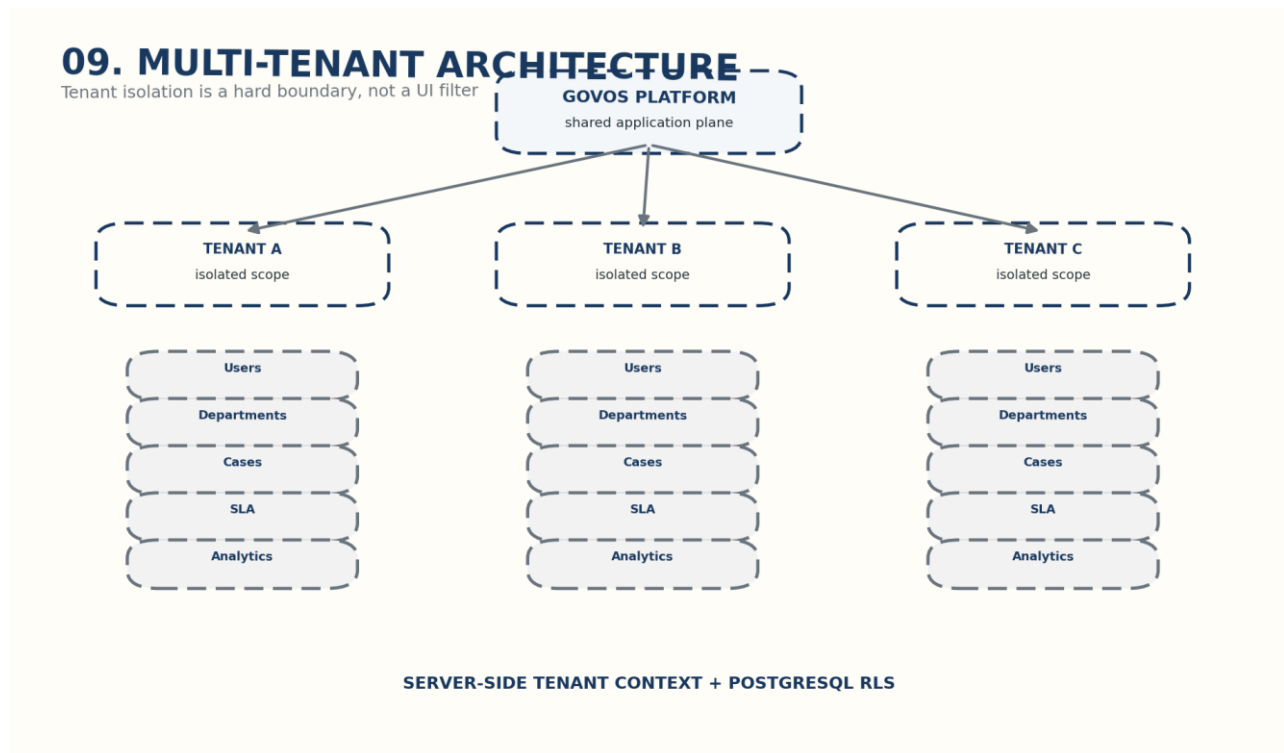
Conceptual and configurable



Conceptual hierarchy; exact government designations are deployment-specific.

Persona	Scope	Core responsibility
SUPER_ADMIN	Platform-wide	Platform governance and tenant administration
TENANT ADMIN	Tenant	Users, departments, configuration and oversight
DEPT HEAD	Department	Queue, workload, performance and exceptions
SUPERVISOR	Team / queue	Assignment, review and escalation
OFFICER / FIELD	Assigned work	Execution, progress, evidence and completion
VERIFIER	Authorized scope	Independent completion verification
CITIZEN	Own cases	Submission, tracking, updates and feedback

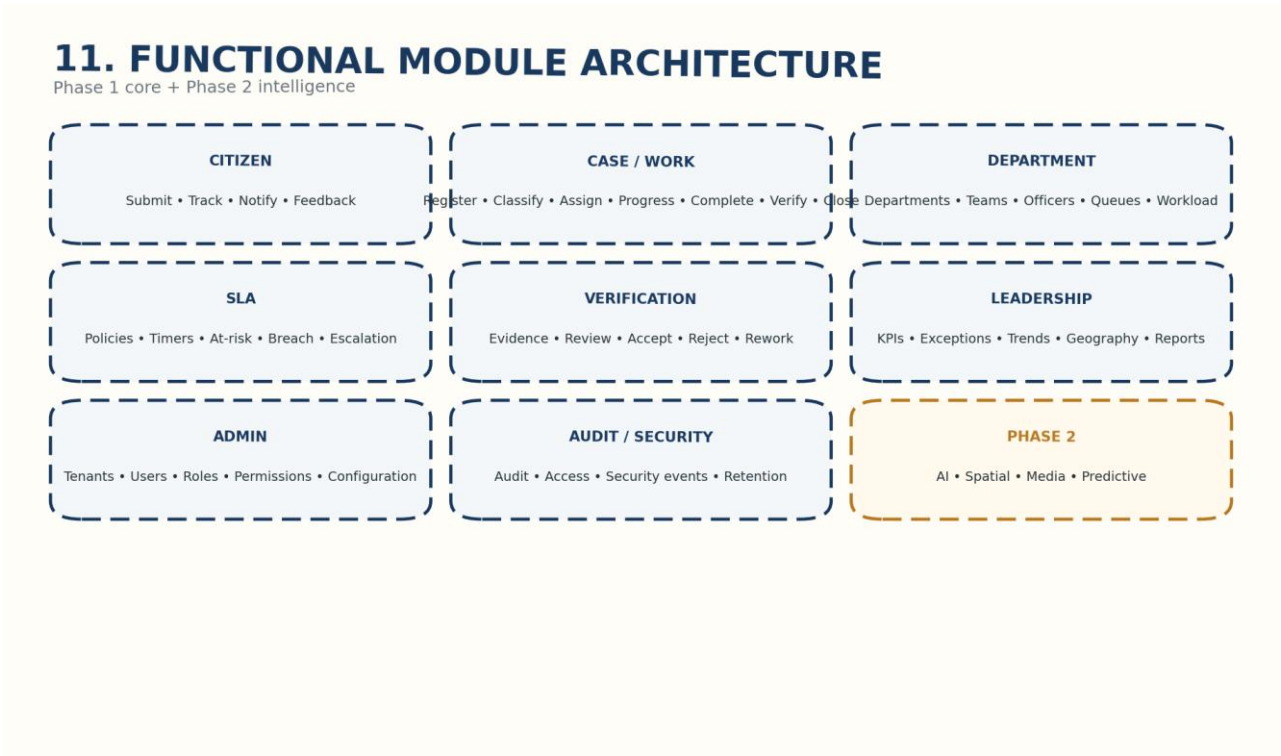
13. Multi-Tenant Architecture



Tenant isolation as a database and application boundary.

- Server-side tenant context must be derived from authenticated identity.
- PostgreSQL RLS should provide a database-level fail-safe.
- Application authorization and database isolation are defense in depth.
- Cross-tenant administrative access must be restricted and auditable.

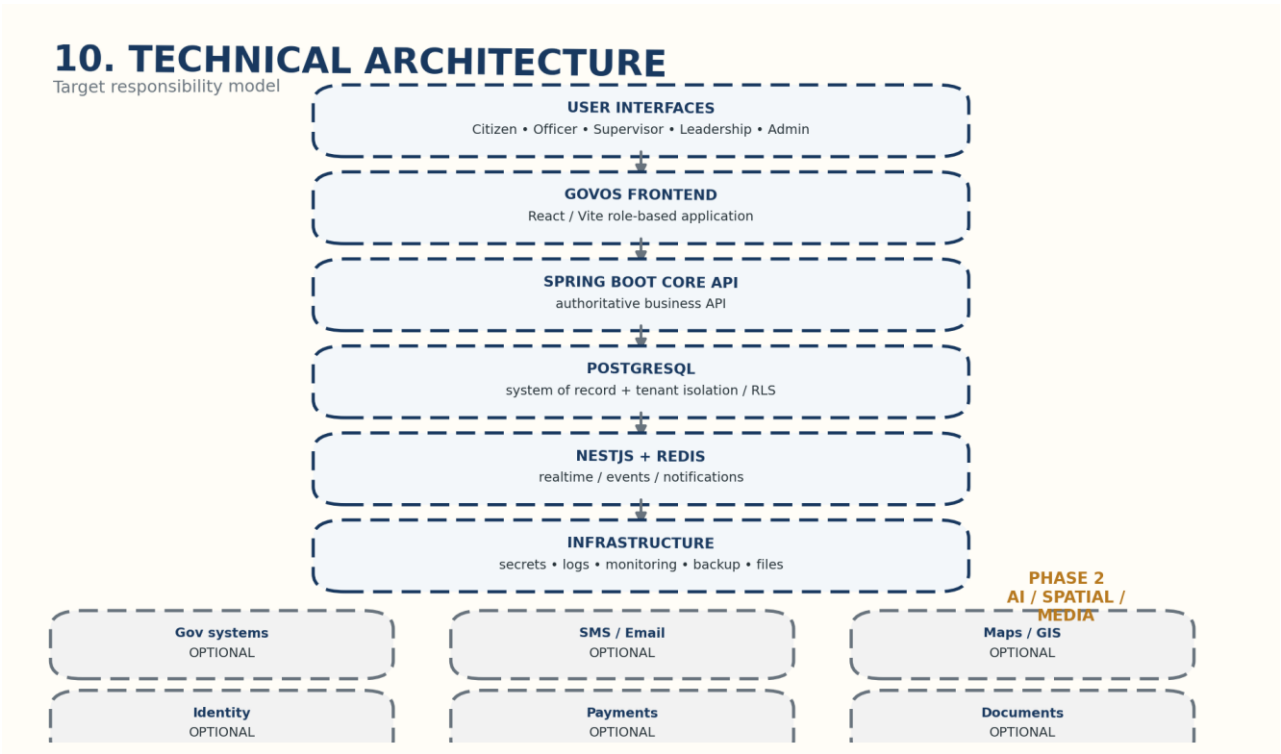
14. Functional Module Architecture



Phase 1 core modules and Phase 2 intelligence boundary.

Module	Phase 1 intent
Citizen Services	Submit, track, notifications, feedback
Case / Work	Registration through closure and reopening
Department	Teams, officers, queues, workload
SLA	Policies, timers, warnings, breach, escalation
Verification	Evidence, review, accept/reject, rework
Leadership	KPIs, exceptions, trends, geography, reports
Administration	Tenants, users, roles, permissions, configuration
Audit / Security	Audit, access, security events, retention
Phase 2	AI, spatial, media and predictive intelligence

15. Technical Architecture



Target responsibility model.

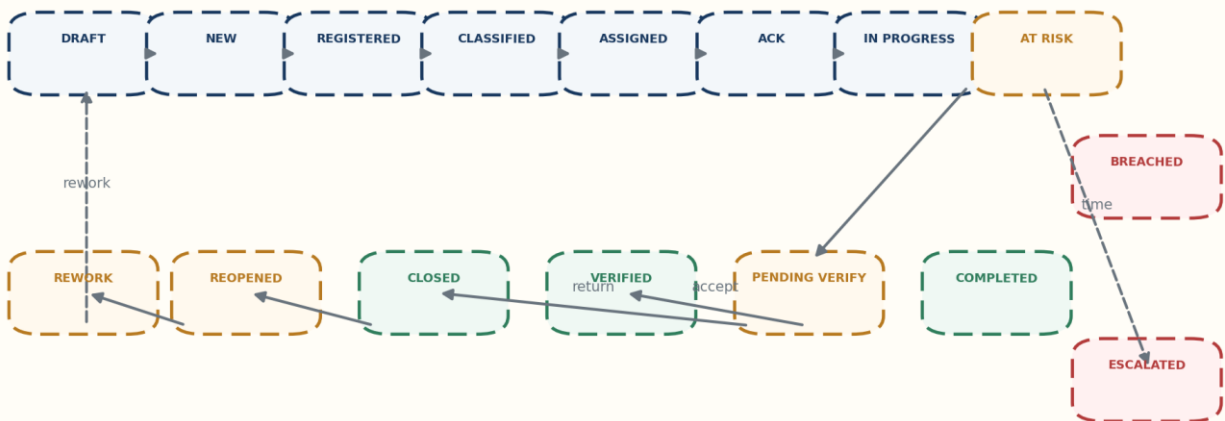
Component	Authoritative responsibility
GovOS Frontend	Presentation and role-based interaction
Spring Boot Core API	Authoritative business rules and APIs
PostgreSQL	System of record and tenant isolation/RLS
NestJS + Redis	Realtime/events/notification event path
Infrastructure	Secrets, logging, monitoring, backup, storage
AI / Spatial	Phase 2 intelligence boundary

Architectural rule: avoid competing business authorities. The current Express backend is a migration fallback, not a second permanent source of business truth.

16. Case / Work Item Lifecycle

12. CASE / WORK ITEM LIFECYCLE

State model — exact permissions are implementation validation items



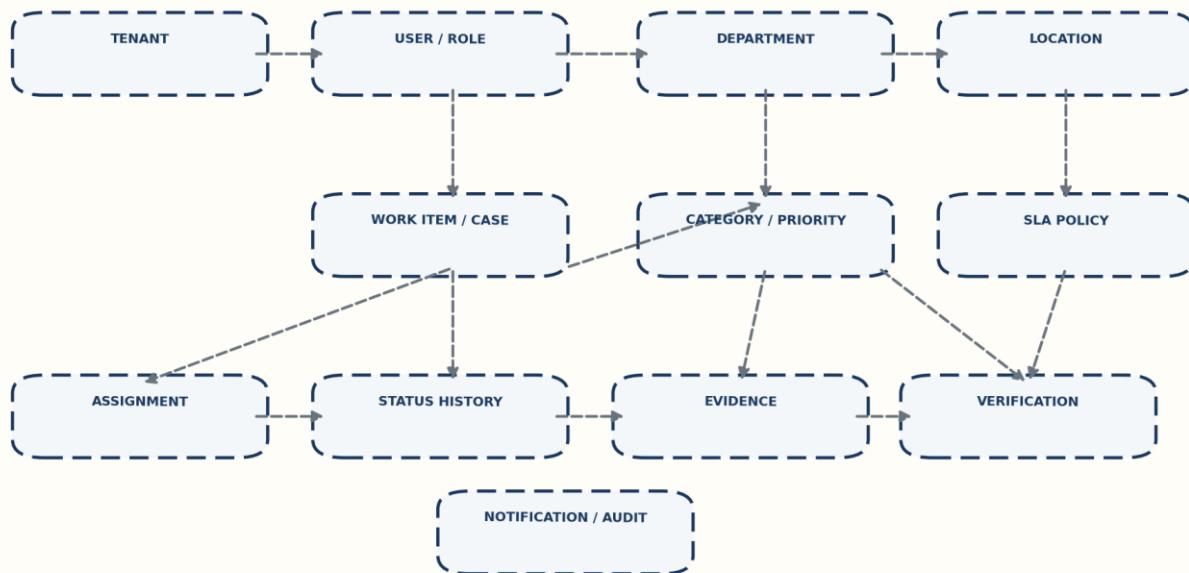
Recommended state model; exact transition permissions require implementation validation.

Transition	Typical initiator	Control
Assignment / reassignment	Supervisor / admin	Scope + audit
Progress / completion	Officer	Assignment scope + audit
At-risk / breach	System	Background worker + audit
Verification	Verifier	Verification authority
Rework	Verifier / supervisor	Reason + new execution
Reopen	Authorized actor	Reason + authorization + audit

17. Conceptual Data Architecture

13. CONCEPTUAL DATA ARCHITECTURE

Business entities and relationships — not a physical schema



Business entity relationship view, not a physical database schema.

Tenant is the isolation root. Work Item / Case is the central operational object. Assignment, status history, SLA, evidence, verification, notification and audit records preserve lifecycle traceability.

18. Audit Architecture

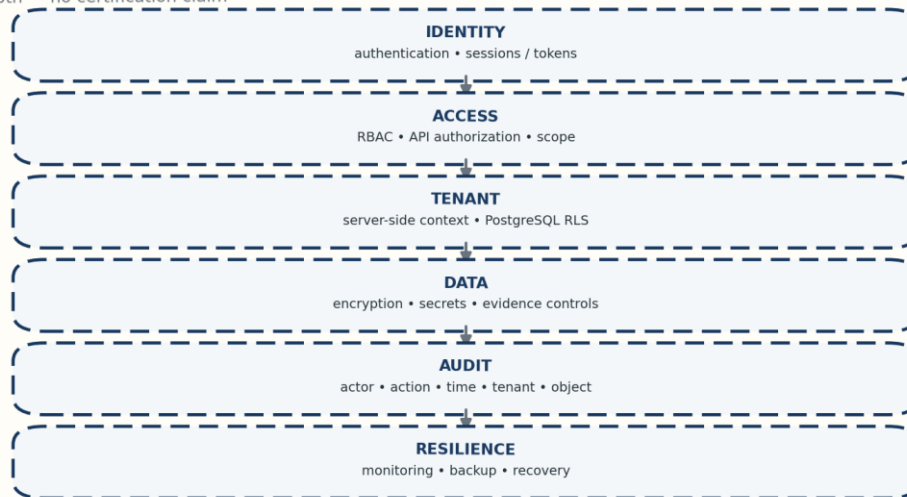
Recommended chain: USER ACTION → AUTHENTICATION → AUTHORIZATION → BUSINESS ACTION → AUDIT EVENT → TIMESTAMP → ACTOR → OBJECT / CASE → OLD VALUE / NEW VALUE where applicable.

Audit element	Purpose
Actor	Who acted
Timestamp	When
Tenant	Administrative boundary
Action	What operation
Object / Case	What was affected
Before / After	What changed
Outcome	Success / failure / exception

19. Security Architecture

14. SECURITY ARCHITECTURE

Defense in depth — no certification claim



BEFORE LIVE CITIZEN DATA → SECURITY / PRIVACY REVIEW + TESTED CONTROLS

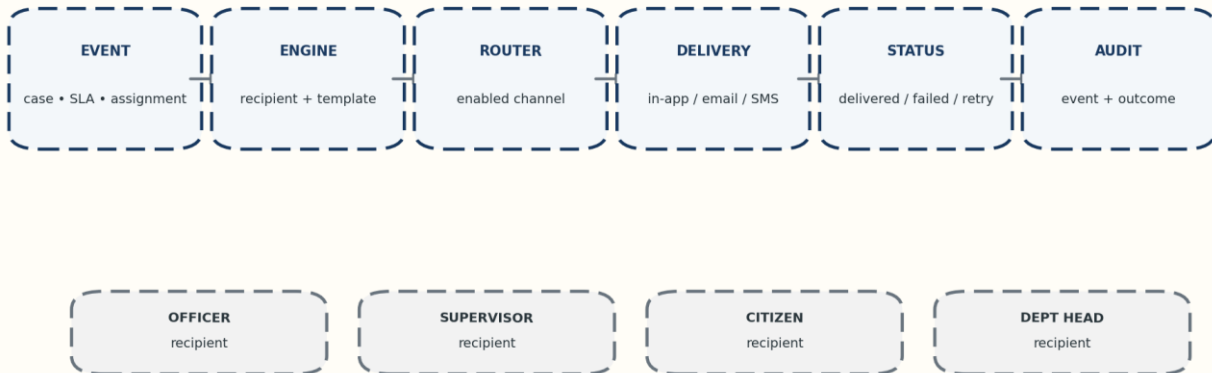
Defense in depth.

- Authentication establishes identity.
- Authorization controls permitted actions and scope.
 - *Tenant isolation must be enforced server-side and at PostgreSQL RLS.*
- Secrets must never be hardcoded.
- Evidence/file access must respect tenant and role scope.
- Independent security/privacy review should precede live citizen data.

20. Notification Architecture

15. NOTIFICATION ARCHITECTURE

Event-driven communication with delivery traceability



Event → recipient → channel → delivery → audit.

SMS, email, messaging and other external channels are configurable integrations; their availability must be confirmed for the deployment.

21. Dashboard Information Architecture

16. DASHBOARD INFORMATION ARCHITECTURE

Role-scoped information, exception-based management



Role-scoped dashboard hierarchy.

22. UI / UX Information Architecture

Persona	Navigation
Citizen	Home → Submit Issue → My Issues → Track → Notifications → Profile
Officer	Dashboard → My Work → Case → Progress → Evidence → Complete
Supervisor	Dashboard → Queue → Assignments → SLA → Escalations → Team
Leadership	Command Centre → Exceptions → Performance → Trends → Geography → Reports
Admin	Users → Roles → Departments → SLA → Categories → Configuration → Audit

23. Mobile / Field Operations

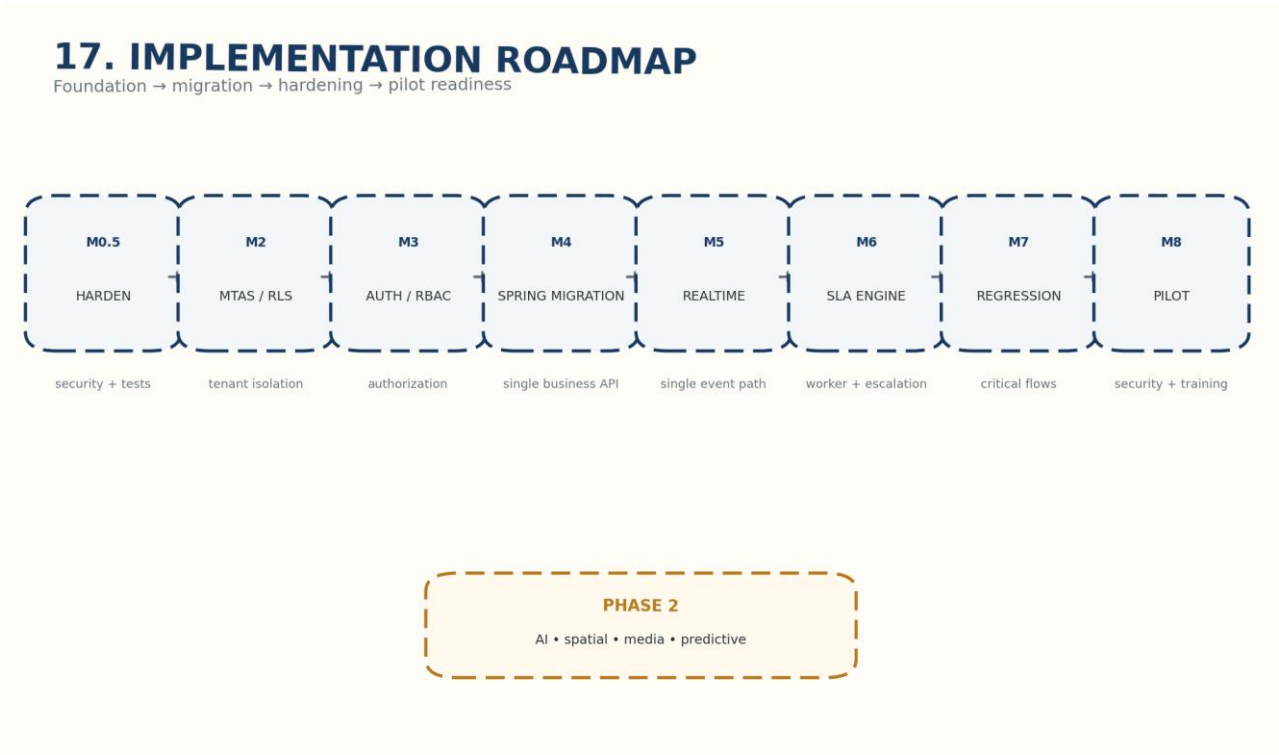
Recommended field sequence: Assigned Work → View Location → Start Work → Capture Progress → Capture Evidence → Complete → Submit for Verification.
GPS, NFC, biometrics and similar device capabilities are not mandatory in this blueprint unless separately approved.

24. Reporting & KPI Model

Report family	Examples	Required KPI metadata
Operational	Open/pending/queues	Definition + source + refresh
SLA	At-risk/breached	Clock + threshold
Department	Performance/workload	Scope + denominator
Officer	Assigned/completed/pending	Assignment scope
Citizen service	Cycle time/status	Population + period
Verification	Verified/rejected/reworked	Decision logic
Audit	Critical actions/access	Event definition
Exception	Stalled/breached/reopened	Exception rule

Every KPI should have a metric name, definition, data source, calculation, frequency and responsible role.

25. Implementation Roadmap



Foundation to pilot-readiness sequence.

Milestone	Focus	Readiness gate
M0.5	Codebase hardening	Security baseline + tests
M2	Multi-tenant DB / RLS	Isolation tests pass
M3	Authentication / RBAC	Authorization tests pass
M4	Express → Spring	Existing workflows verified on target API
M5	Realtime	Single event path
M6	SLA engine	Warnings/breach/escalation verified
M7	Regression	Critical workflows pass
M8	Pilot readiness	Security + deployment + training + governance

26. MVP Scope

Priority	Scope
MUST HAVE	Authentication, RBAC, tenant isolation/RLS, citizen issue submission, case lifecycle, assignment, SLA, escalation, completion, verification, closure/reopen, audit, notifications, basic dashboards
SHOULD HAVE	Department workload, GIS foundation, assets, projects, documents/Peshi, richer analytics, field enhancements
FUTURE	AI, media intelligence, predictive governance and advanced intelligence

27. Phase 2 — Intelligence

Phase 2 remains deliberately separated from the Phase 1 pilot: AI-assisted intelligence, GIS/spatial intelligence, media intelligence, advanced analytics and predictive capabilities.

The source explicitly parks real-world video reporting, photo evidence capture, AI issue categorization/media analysis, universal issue intelligence, observation intelligence and predictive governance outside Phase 1.

28. Risks & Controls

Risk	Control
Split-brain backend	Migrate business logic to Spring Boot; keep Express only as controlled fallback.
Cross-tenant leakage	PostgreSQL RLS + server-side tenant context + tests.
Weak authentication	Production OTP/JWT + secret management + abuse/expiry tests.
Passive SLA	Background worker + tested escalation.
Unverified completion	Separate completion and verification.
Duplicate business authorities	Explicit component responsibility model.
Integration dependency	Document system authority and integration boundaries.
Security/privacy	Independent review before live citizen data.
AI scope creep	Phase 2 approval gate.
Weak pilot evidence	Baseline + KPIs + acceptance criteria + evaluation.

29. Final Architecture Principles

Principle	Meaning	Government value
Single ownership	Every work item has an accountable owner.	Accountability
Time-bound execution	SLA and escalation are explicit.	Service discipline
Verified outcomes	Completion and verification are distinct.	Credible reporting
Auditability	Critical actions are recorded.	Traceability
Tenant isolation	Municipal data is separated by design.	Data protection
Exception management	Leadership focuses on risk and delay.	Better decisions
Interoperability	GovOS coexists with authoritative systems.	Avoid duplication
Phase discipline	AI/media intelligence stays outside Phase 1.	Scope control

30. Final Consistency Checklist

- Citizen submission enters the same case/work lifecycle as internal operations.
- Every case has explicit ownership and an auditable assignment record.
- Reassignment is controlled and auditable.
- SLA monitoring is connected to execution and breach escalation.
- Completion is distinct from verification.
- Verification can return work for rework.
- Closed work can be reopened through a controlled transition.
- Leadership reporting distinguishes completed from verified outcomes.
 - *Tenant isolation exists at application and database layers.*
- PostgreSQL remains the system of record.
- Spring Boot remains the authoritative business API.
- NestJS is limited to realtime/event responsibility.
- AI/spatial/media intelligence remains Phase 2.
- External integrations are labelled optional unless confirmed.
- Assumptions requiring government/business validation are not presented as established facts.

Source Discipline & Approval Note

SOURCE-DERIVED = supported by the attached GovOS blueprint. RECOMMENDED = architecture/product recommendation. ASSUMPTION / VALIDATION REQUIRED = must be confirmed by the pilot authority, business owner or implementation team.

This document is a professional project blueprint and planning reference. It is not government approval, procurement authorization, a legal certification or a final departmental SOP.